CIE AS PHYSICS (9702) • QUESTION PAPER

8.4 The Diffraction Grating

Syllabus Section: 8. Superposition **Paper Type:** Multiple Choice (Paper 1) **Total:** 16 Questions (16 Marks)

Question 1 Easy [1 mark]

A diffraction pattern is observed when monochromatic light is incident on a diffraction grating.

Which row describes the effect of replacing the grating with one that has more lines per metre?

	Number of orders of diffraction visible	Angle between first and second orders of diffraction
A	increases	increases
B	increases	decreases
C	decreases	increases
D	decreases	decreases

A

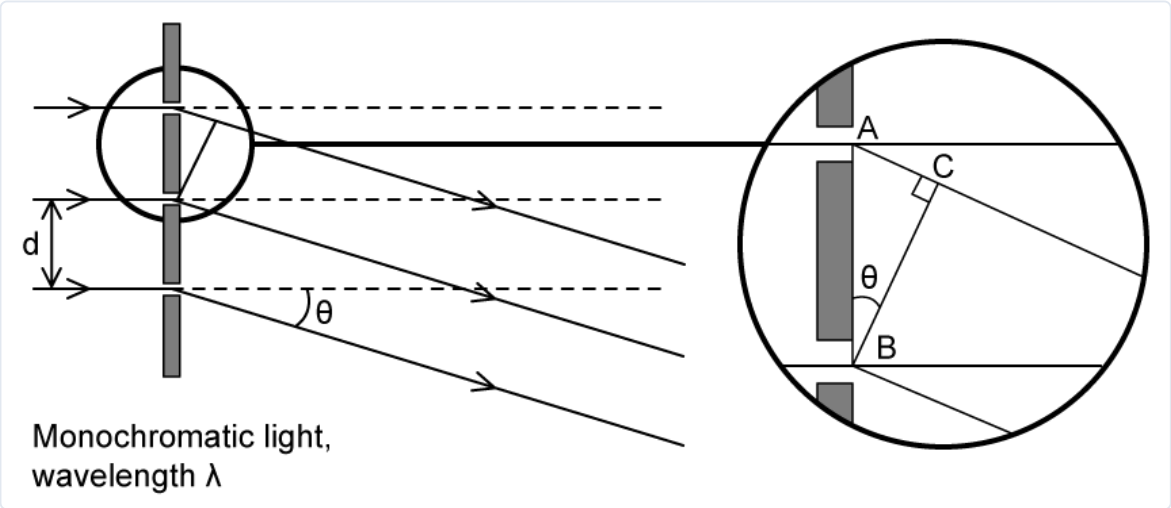
B

C

D

Question 2 Easy [1 mark]

The diagram shows a diffraction grating and a monochromatic source of light with wavelength (λ).



What happens to the angle of diffraction (θ) when the wavelength of light is increased, but all the other equipment remains the same?

A The angle of diffraction remains the same

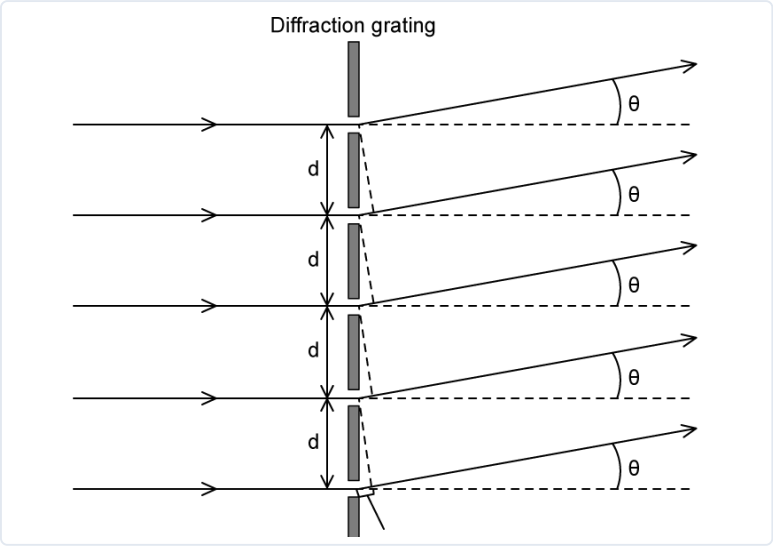
B The angle of diffraction is decreased

C The angle of diffraction is increased

D There is not enough information to know what will happen

Question 3 Easy [1 mark]

The diagram shows a diffraction grating with a slit distance (*d*) and a monochromatic source of light incident on the grating.



What happens to the angle of diffraction when the distance between the slits on the grating is reduced?

A The angle of diffraction remains the same

B The angle of diffraction is decreased

C The angle of diffraction is increased

D There is not enough information to know what will happen

Question 4 Easy [1 mark]

Monochromatic light incident on a diffraction grating creates both primary and secondary maxima in the diffraction pattern.

What is the relationship between the number of slits in the diffraction grating, when the number of slits is less than 20, and the number of secondary maxima that can be observed on the screen?

A The number of primary and secondary maxima in each diffraction pattern is the same

B The number of secondary maxima in the diffraction pattern is always one less than the number of primary maxima.

C The number of secondary maxima in the diffraction pattern is always the same as the number of slits in the diffraction grating

D The number of secondary maxima between the primary maxima is always two less than the number of slits in the diffraction grating

Question 5 Easy [1 mark]

What is a diffraction grating?

A Two slits created as two gaps within a material through which waves can be diffracted

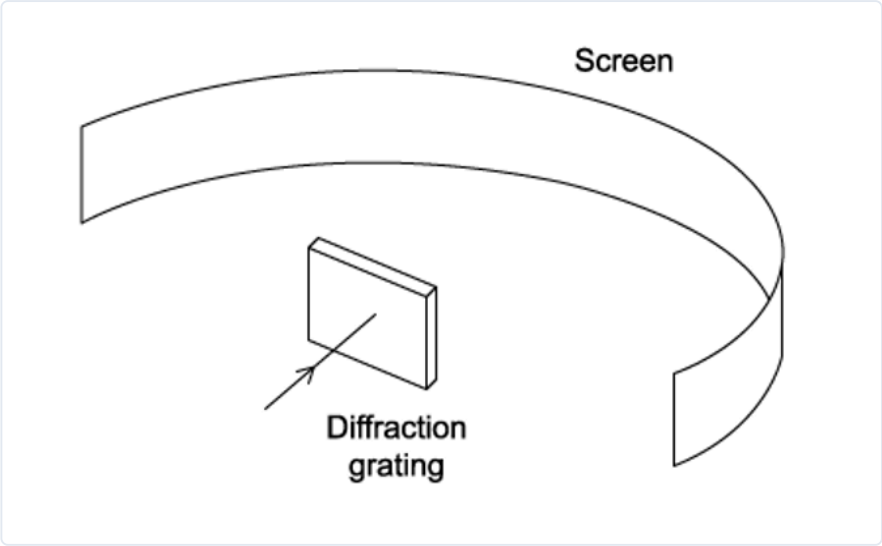
B A plate consisting of a very large number of parallel, identical, closely-spaced slits

C A circular aperture through which light can be diffracted and then images resolved

D A photographic filter that allows light in only one plane to pass through

Question 6 Medium [1 mark]

A beam of monochromatic light with a wavelength of 690 nm is directed at a diffraction grating with 300 lines per mm as shown in the diagram



This set up produces a series of maxima on the screen, what is the greatest number of maxima that can be observed?

A 9

B 8

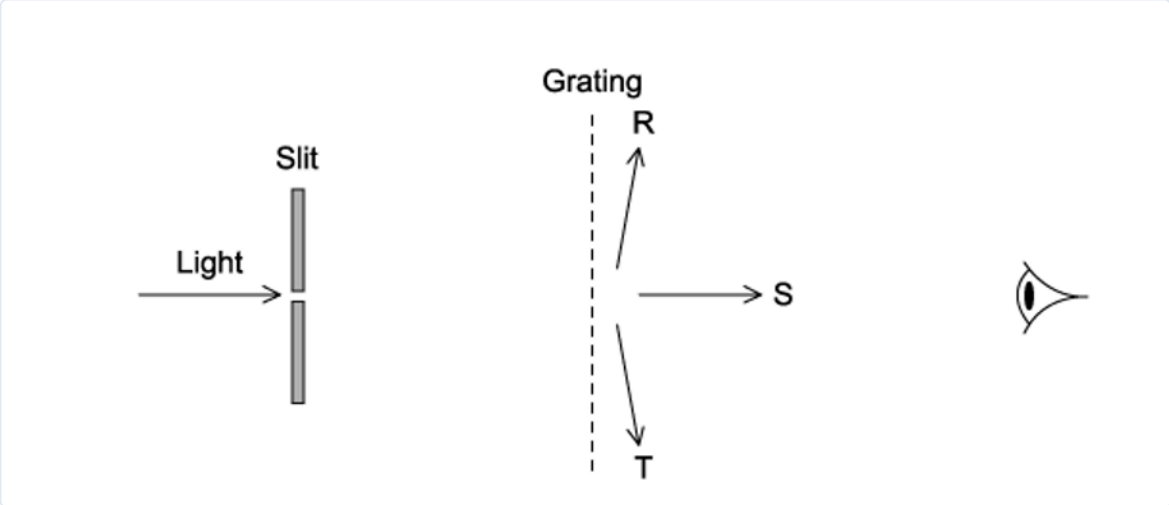
C 5

D 4

Question 7 Medium [1 mark]

Monochromatic light with a wavelength of 600 nm is observed through a diffraction grating with 500 lines per mm.

The light is first passed through a narrow slit where the lines from the slit are parallel to the lines in the grating.



The observer changes position to; R – parallel with the grating, S – opposite the slit, T – parallel to the grating on the opposite side.

What is the total number of images they would see from R, S and T combined?

A 7

B 6

C 4

D 3

Question 8 Medium [1 mark]

A student set up an experiment with a diffraction grating. They put a narrow beam of monochromatic light incident along the normal. They found that the third-order diffracted beams are formed at angles of 45° to the original direction.

Which of the following gives the highest order of diffracted beam produced by this grating ?

A 6th

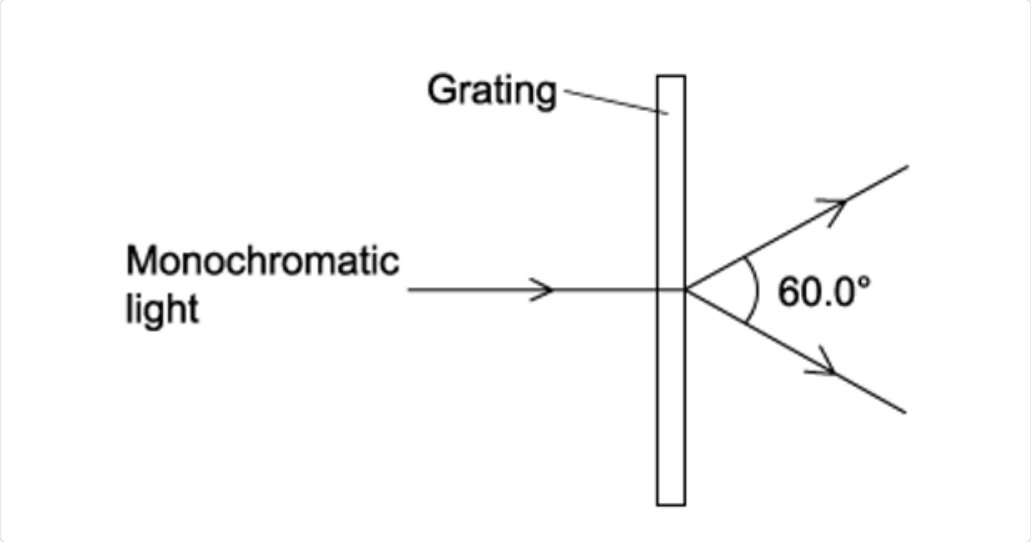
B 5th

C 4th

D 3rd

Question 9 Hard [1 mark]

The wavelength of monochromatic light can be determined using a diffraction grating. This is shown in the diagram.



The angle between the first-order diffraction maxima is 60°, and the spacing of the slits is 1.15×10^{-6} m.

What is the wavelength of the light?

A 575 nm

B 498 nm

C 996 nm

D 287 nm

Question 10 Hard [1 mark]

When a parallel beam of white light is incident to normally on a diffraction grating the second-order and third-order spectra partially overlap.

When the second order wavelength is 600 nm, which of the following third-order wavelengths would appear at the same angle?

A 900 nm

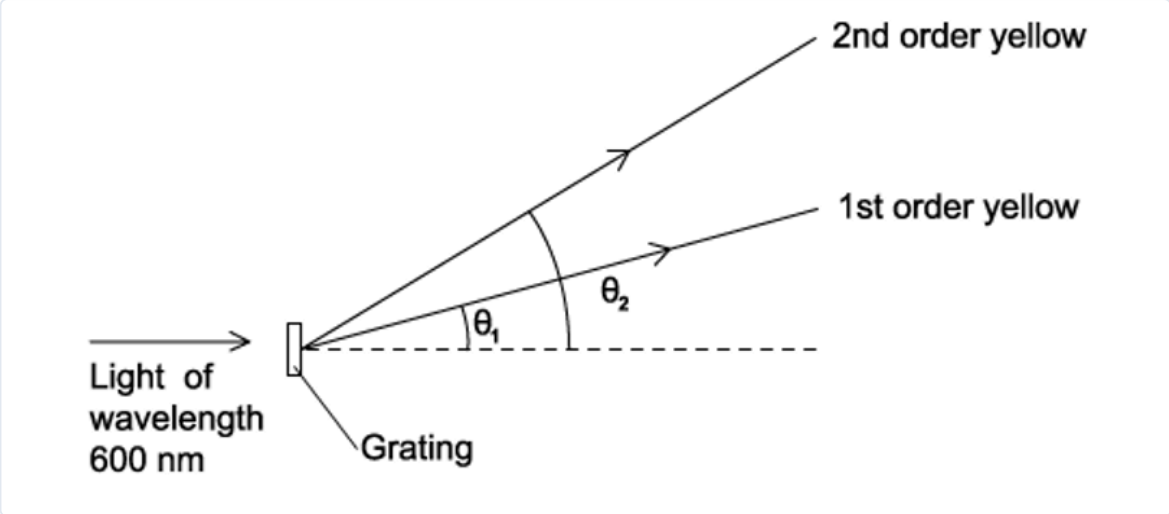
B 600 nm

C 400 nm

D 300 nm

Question 11 Hard [1 mark]

Yellow light with a wavelength of 600 nm is used in a diffraction grating experiment. The grating has a slit separation of 2.00 μm.



What would be the angular separation between the first order and second order maxima?

A 54.3°

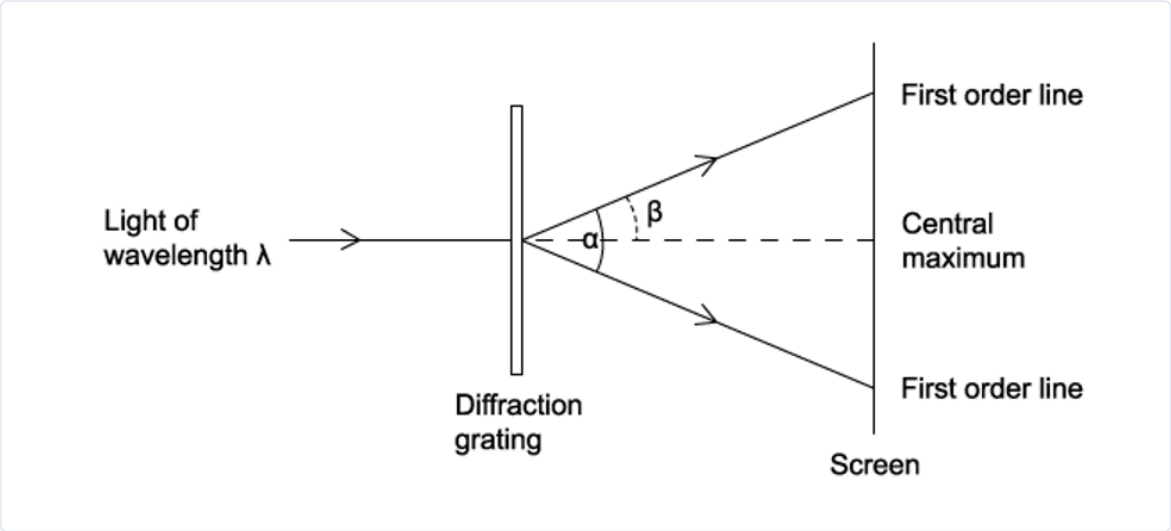
B 36.9°

C 19.4°

D 17.5°

Question 12 Hard [1 mark]

An experiment was set up to investigate light passing through a diffraction grating with slit spacing *d*. The fringe pattern was observed on a screen.



Which of the following gives the angle between the two first-order lines?

A $2 \quad sin^{-1} \frac{\lambda}{2a}$

B $2 \quad \sin^{-1} \frac{\lambda}{a}$

C $\sin^{-1} \frac{\lambda}{2a}$

D $\sin^{-1} \frac{\lambda}{a}$

Question 13 Hard [1 mark]

A student carried out an experiment on monochromatic light with a wavelength of 5.30×10^{-7} m. They set up the source so that it was incident normally to the diffraction grating.

The student observed that the first-order maximum was 15.4° to the direction of the incident light.

What would be the angle between the first order and second order maxima?

A 7.6°

B 15.4°

C 16.7°

D 32.1°

Question 14 Hard [1 mark]

A monochromatic light beam is incident normally to a diffraction grating and nine maximas are observed on the screen.

What is the range for the angle θ between the central and first-order maxima?

A $5.74° \leq \theta \leq 6.38°$

B $9.59° < \theta \leq 11.5°$

C $11.5° < \theta \leq 14.5°$

D $11.5° < \theta < 53.1°$

Question 15 Hard [1 mark]

Fig. 1.1 shows a monochromatic light beam which is incident normally to a diffraction grating. The angle between the central and second-order maxima is 18°.

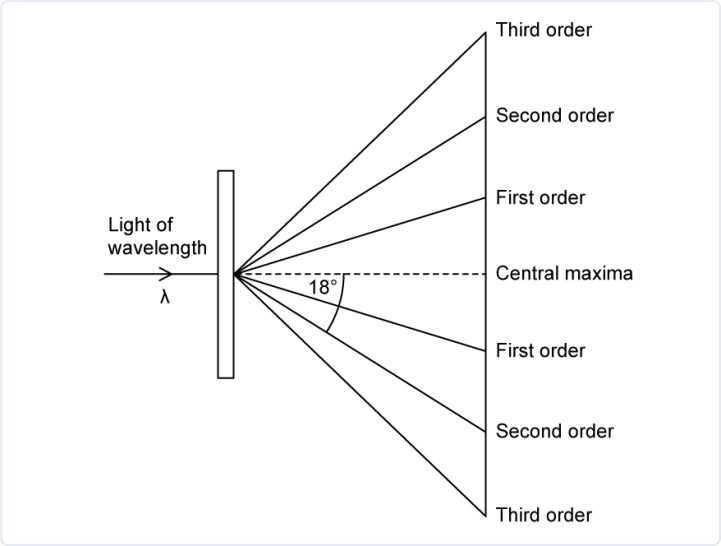


Fig. 1.1

What is the angle between the two third-order maxima lines?

A 27.6°

B 36.0°

C 54.0°

D 55.2°

Question 16 Hard [1 mark]

A physics student is investigating the properties of a diffraction grating using laser light with a wavelength of 6.20×10^{-7} m. The laser beam is directed perpendicularly onto the grating surface.

During the experiment, the student measures the second-order maximum at an angle of $32°$ from the normal to the grating.

What is the angle between the second-order and third-order maxima?

A 12°

B 21°

C 32°

D 53°